Notes: Hirshleifer, Lim, and Teoh (JF, 2009)

Driven to Distraction: Extraneous Events and Underreaction to Earnings News

Contents

1 Big picture

- **Question.** Do *extraneous* news events—specifically, same-day earnings announcements by other firms—distract investors and make them underreact to a firm's own earnings news?
- **Setting.** The paper studies U.S. quarterly earnings announcements from 1995 to 2004 using CRSP-Compustat and I/B/E/S data.
- **Core challenge.** A positive relation between "busy" announcement days and weaker market reactions could reflect true distraction, but it could also come from calendar seasonality, reporting-timing patterns, date errors, information leakage, or the possibility that different *types* of firms announce on crowded days.
- **Main conceptual contribution.** The paper turns limited attention into a direct empirical test by measuring investors' *information load.* Instead of using day-of-week or market-volume proxies for inattention, it counts how much other same-day public news competes for attention.
- **Investor distraction hypothesis.** More competing announcements should lead to:
  - a weaker immediate price reaction to the firm's earnings surprise,
  - weaker announcement-window trading volume,
  - and stronger post-earnings-announcement drift.
- **Main empirical design.** In each quarter, the authors sort earnings announcements by the firm's own earnings surprise and by the number of same-day announcements by *other* firms. They then study announcement returns, post-announcement drift, and trading volume, and also estimate multivariate regressions with rich controls.
- **Main univariate result.** On low-news days, the spread in announcement-window abnormal returns between good-news and bad-news firms is 7.02%; on high-news days it is only 5.81%. For post-announcement drift, the spread is 2.66% on low-news days and 7.18% on high-news days.
- **Main multivariate result.** In regressions with controls, the interaction between earnings surprise and the number of competing announcements is negative for announcement returns and positive for post-announcement drift. Economically, immediate sensitivity to earnings news is about 13.3% lower on high-news days than on low-news days.
- **Main timing result.** The extra drift caused by distraction does not show up right away. It becomes significant around 45 trading days after the announcement, is strongest around 60 days, and fades by 90 days.
- **Main trading-strategy result.** A drift strategy that is long good-news firms and short bad-news firms earns a Fama–French three-factor alpha of 1.64% per month after high-news-day announcements, versus only 0.77% and statistically insignificant after low-news-day announcements.
- **Main comparative-statics result.** Unrelated announcements distract more than related announcements, and announcements with big absolute earnings surprises distract more than those with small surprises.
- **Main asymmetry result.** The evidence is mixed but suggestive that distraction is somewhat stronger for small firms and for positive earnings surprises.

- **Economic takeaway.** Peer news can crowd out attention to a firm's own earnings announcement. The paper therefore gives a micro-foundation for why underreaction and post-announcement drift can intensify when the market is flooded with competing information.

2 Data and measurement (key objects)

2.1 Data sources and sample window

The paper combines quarterly earnings announcement data from the CRSP-Compustat merged database with analyst forecast data from I/B/E/S. The sample period runs from 1995 to 2004.

To compute the number of daily quarterly earnings announcements, the authors use Compustat announcement dates for all firms. When an announcement date is available in both Compustat and I/B/E/S but differs across the two sources, they take the earlier date, following DellaVigna and Pollet's validation exercise for the post-1994 period.

Interpretation. The design needs *two* things at once:

- an accurate earnings surprise measure for the test firm, which is why I/B/E/S coverage matters;
- and a broad count of all same-day competing announcements, which is why the authors do *not* limit the count to I/B/E/S firms only.

2.2 Earnings surprise

For firm i in fiscal quarter q, the earnings surprise is

$$FE_{iq} = \frac{e_{iq} - F_{iq}}{P_{iq}}, \tag{1}$$

where e_{iq} is announced earnings from I/B/E/S, F_{iq} is the consensus analyst forecast (the median of the most recent analyst forecasts), and P_{iq} is the stock price at the end of the corresponding quarter.

The consensus forecast uses one- or two-quarter-ahead forecasts issued or reviewed within the last 60 calendar days before the earnings announcement. If an analyst has multiple valid forecasts, only the most recent one is kept.

Interpretation. The surprise measure is standardized by stock price, not by forecast dispersion or earnings level. This makes the variable comparable across firms and quarters.

2.3 Competing-news measure

The key distraction proxy is the total number of quarterly earnings announcements by *other firms* on the same calendar day. In each calendar quarter, all announcements are sorted into deciles by this daily count. Let $NRANK$ denote the resulting decile rank:

- $NRANK = 1$: low-news days,
- $NRANK = 10$: high-news days.

Table I shows the distribution of the raw daily count:

- mean: 120.8,
- standard deviation: 129.7,
- median: 71,
- 90th percentile: 290.

Interpretation. This is a *big* state variable. Some earnings announcements occur on very quiet days, others on days with several hundred competing announcements.

2.4 Announcement and post-announcement abnormal returns

The paper measures abnormal performance using buy-and-hold abnormal returns relative to size and book-to-market matched portfolios.

For announcement date t, the 2-day announcement-window abnormal return is

$$CAR[0,1]_{iq} = \prod_{k=t}^{t+1}(1+R_{ik}) - \prod_{k=t}^{t+1}(1+R_{pk}), \tag{2}$$

where R_{ik} is the return on firm i and R_{pk} is the return on the matching size–B/M portfolio.

The main post-announcement drift window is

$$CAR[2,61]_{iq} = \prod_{k=t+2}^{t+61}(1+R_{ik}) - \prod_{k=t+2}^{t+61}(1+R_{pk}). \tag{3}$$

The paper focuses on a 60-trading-day horizon because Bernard and Thomas show that most post-earnings-announcement drift occurs over roughly this interval.

Interpretation. The design cleanly separates the *immediate* reaction from the *delayed* correction. If distraction is real, the first should weaken and the second should strengthen.

2.5 Why the paper uses earnings-surprise deciles

The relation between announcement returns and raw forecast errors is strongly nonlinear, especially for small negative surprises. To reduce outlier influence and linearize the empirical relation, the paper works with the decile rank of earnings surprise rather than the raw value of FE.

Interpretation. This is not cosmetic. The whole interaction design becomes easier to interpret when moving one decile in earnings surprise has an approximately linear return effect.

2.6 Control variables

The multivariate tests control for variables known to affect the reaction to earnings news:

- firm size,

- book-to-market decile,
- $\log(1 + \#\text{Analysts})$,
- reporting lag (plus its square and cube),
- institutional ownership,
- earnings persistence,
- earnings volatility,
- share turnover,
- year, month, day-of-week, and Fama–French 10-industry fixed effects.

The paper defines:

- **Earnings persistence** as the first-order autocorrelation of quarterly earnings per share over the previous four years;
- **Institutional ownership** as the percentage of shares held by institutions at the end of the most recent quarter using CDA/Spectrum 13F data;
- **Earnings volatility** as the four-year standard deviation of deviations of quarterly earnings from earnings one year earlier;
- **Reporting lag** as the number of days between fiscal-quarter end and the earnings announcement;
- **Share turnover** as average monthly share trading volume divided by average shares outstanding over the prior year.

Interpretation. These controls matter because busy announcement days are not random. High-news days tend to be associated with larger firms, more analyst following, higher institutional ownership, shorter reporting lags, and more persistent earnings.

2.7 Abnormal trading volume

To measure whether distraction weakens *attention-related trading*, the paper defines abnormal volume on day j relative to announcement date t as

$$VOL[j] = \log(\text{DollarVol}_{t+j} + 1) - \frac{1}{30} \sum_{k=t-41}^{t-11} \log(\text{DollarVol}_k + 1). \tag{4}$$

The 2-day abnormal trading-volume response is then

$$VOL[0,1] = \frac{VOL[0] + VOL[1]}{2}. \tag{5}$$

Interpretation. If competing news really distracts investors, it should reduce not only prices' sensitivity to earnings news but also the abnormal trading response around the announcement.

2.8 Descriptive facts that matter later

A few patterns in Table I are especially important for the rest of the paper:

- high-news-day announcements come from larger firms on average,
- high-news-day firms have greater institutional ownership and analyst following,
- earnings surprises on high-news days are less negative on average,
- high-news-day firms have more persistent earnings and shorter reporting lags,
- the relation between characteristics and $NRANK$ is not monotonic across all deciles.

Reporting lag is the strongest standardized predictor of $NRANK$ in the table's regression analysis.

Interpretation. The nonmonotonicity is why the paper leans heavily on multivariate controls instead of relying only on raw sorts.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and predictions

The paper's central mechanism is simple. Investors have limited attention. When many other firms also announce earnings on the same day, investors face a larger information load. That makes them slower to process the focal firm's earnings news.

This yields three predictions:

1. immediate price reactions should be weaker on high-news days;
2. post-announcement drift should be stronger on high-news days;
3. trading-volume reactions should also be weaker on high-news days.

Interpretation. The power of the design is that distraction should have *opposite* effects on the announcement window and the post-announcement window. That makes many alternative explanations harder to sustain.

3.2 Univariate double sorts

In each calendar quarter, the paper independently sorts all earnings announcements into a 10×10 grid by:

- the firm's own earnings surprise decile (FE), and
- the same-day announcement-count decile ($NRANK$).

The spread $FE10 - FE1$ in $CAR[0, 1]$ measures how strongly prices react immediately to earnings news. The spread $FE10 - FE1$ in $CAR[2, 61]$ measures how much underreaction remains to be corrected afterwards.

For the most extreme comparison, the paper also estimates

$$CAR = a_0 + a_1 FE10 + a_2 NRANK10 + a_3(FE10 \times NRANK10) + \varepsilon, \tag{6}$$

where $FE10$ indicates top-decile earnings news and $NRANK10$ indicates top-decile announcement crowding. The coefficient a_3 tests whether the good-news minus bad-news spread differs between high-news and low-news days.

Interpretation. This is a difference-in-differences design in the simplest possible form.

3.3 Baseline multivariate specification

The main regression is

$$CAR = a_0 + a_1 FE + a_2 NRANK + a_3(FE \times NRANK) + \sum_{i=1}^{n} c_i X_i + \sum_{i=1}^{n} b_i(FE \times X_i) + \varepsilon. \tag{7}$$

The dependent variable is either $CAR[0, 1]$ or $CAR[2, 61]$. The interaction term $FE \times NRANK$ is the key coefficient:

- for $CAR[0, 1]$, distraction predicts $a_3 < 0$;
- for $CAR[2, 61]$, distraction predicts $a_3 > 0$.

Standard errors are adjusted for heteroskedasticity and clustered by the day of announcement.

Interpretation. This regression asks whether the slope of returns with respect to earnings news becomes flatter immediately, but steeper later, when more other announcements compete for attention.

3.4 What the main coefficients mean

With controls included, Table III reports:

- $FE \times NRANK = -0.015$ for $CAR[0, 1]$,
- $FE \times NRANK = +0.049$ for $CAR[2, 61]$.

Using the fitted coefficients, the sensitivity of announcement returns to earnings news is about 1.013 on low-news days ($NRANK = 1$) and 0.878 on high-news days ($NRANK = 10$), a decline of roughly 13.3%.

For post-announcement drift, the same coefficients imply substantially larger delayed reactions on high-news days than on low-news days. Without controls the paper reports a 75.4% larger drift sensitivity on high-news days; with controls the increase is 42.9%.

Interpretation. The regressions say the same thing as the raw tables, but more cleanly: crowded-news days do not just look different in levels; they change the *slope* linking returns to earnings news.

3.5 Timing of the drift correction

The paper repeats the post-announcement regression over alternative horizons:

- $CAR[2, 30]$,
- $CAR[2, 45]$,
- $CAR[2, 61]$,
- $CAR[2, 75]$,
- $CAR[2, 90]$,
- and $CAR[3, 61]$ to address date-error concerns.

Table IV shows that the distraction effect is insignificant over the first 30 trading days, becomes significant around 45 trading days, is strongest around 60 trading days, and then weakens by 90 days.

Interpretation. This pattern matters. It looks like gradual price correction, not a mechanical announcement-date artifact.

3.6 Volume regression

Because both positive and negative extreme earnings surprises can trigger heavy trading, the paper uses the decile rank of *absolute* earnings surprises (AFE) in the volume regression:

$$VOL[0, 1] = \alpha_0 + \alpha_1 AFE + \alpha_2 NRANK + \text{controls} + \varepsilon. \tag{8}$$

The regressions also include market-wide abnormal trading volume over the same two days.

Table VI shows that the coefficient on $NRANK$ is negative and highly significant in every specification, around -0.020 to -0.026.

Interpretation. Crowded days dampen the *trading response* as well as the price response. This strengthens the claim that the key object is attention, not just return dynamics.

3.7 Trading-strategy specification

At the end of each month from March 1995 to December 2004, stocks are independently sorted into 5×5 portfolios based on:

- the most recent quarterly earnings surprise in the last three months, and
- the number of announcements on the day of that earnings announcement.

Within each announcement-count quintile, the paper forms a hedge portfolio that is long good-news firms ($FE = 5$) and short bad-news firms ($FE = 1$), then estimates Fama–French three-factor alphas.

Interpretation. If distraction intensifies post-earnings-announcement drift, a PEAD strategy should work especially well after high-news-day announcements.

3.8 Comparative statics and heterogeneity

The paper also asks *which* competing announcements are most distracting and *which* firms are most vulnerable.

It splits competing announcements three ways:

1. related versus unrelated industry news,
2. big versus small absolute earnings surprises,
3. announcements by small versus large firms.

It also estimates whether distraction differs by:

- small versus large announcing firms,
- positive versus negative earnings surprises.

For the positive-versus-negative test, the paper estimates a piecewise-linear specification

$$CAR = a_0(X) + a_1(X)FE + a_2(X)FE_p + \varepsilon, \tag{9}$$

where $FE_p = \max(FE - 6, 0)$ allows the earnings-news slope and the distraction effect to differ for positive surprises.

Interpretation. The paper is not content with showing an average effect. It asks what sort of competing news truly diverts attention, and whether some firms or some kinds of surprises are more exposed.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

There is no formal instrumental-variable strategy in the baseline announcement regressions. Identification comes from cross-sectional and time-series variation in the number of other firms' same-day earnings announcements, after controlling for:

- firm characteristics,
- industry effects,
- month, year, and day-of-week effects,
- and nonlinear reporting-lag effects.

The paper is therefore identified off differences in how crowded an announcement day is, conditional on observable determinants of earnings reactions.

Interpretation. The design is credible if same-day announcement crowding captures competition for investor attention rather than some omitted confound that also directly changes the earnings-response slope.

4.2 Why reverse causality is weak

The focal firm's own immediate return reaction cannot plausibly determine how many *other* firms announce on the same day. The main problem is not reverse causality but omitted-variable bias:

crowded announcement days may systematically differ in other ways.

Interpretation. This is why the paper spends so much effort on calendar controls, same-firm comparisons, and placebo tests.

4.3 Date errors and announcement leakage

A concern is that high-news days might contain more measurement error in announcement dates, which would mechanically weaken announcement-window return sensitivity.

The paper checks announcement dates against the hand-collected Lexis-Nexis / PR-newswire dates from DellaVigna and Pollet. It also shifts the return windows to start earlier or later and finds similar results.

A second concern is that high-news days may be associated with more preannouncement leakage. Figure 3 and the accompanying regressions show no significant difference between high-news and low-news days in preannouncement return sensitivity to earnings surprises.

Interpretation. If the effect were just bad timestamping or earlier leakage, the paper should have found it before day 0. It does not.

4.4 Calendar seasonality and placebo controls

The number of announcements is highly seasonal. The authors therefore include year, month, and weekday fixed effects. More importantly, they build placebo measures from the number of announcements on similar dates in the *previous* year:

- $LNRANK1$: the closest calendar day with the same weekday,
- $LNRANK2$: the same weekday, same month, and same week-of-month in the previous year.

Table V shows that these placebo ranks can predict weaker immediate reactions and stronger drift when used alone, but become insignificant once the current year's actual $NRANK$ is included. By contrast, actual $NRANK$ stays significant.

Interpretation. Seasonality matters, but it is not the whole story. The current level of same-day crowding contains incremental explanatory power beyond calendar regularities.

4.5 Same-firm high-news versus low-news comparisons

Another worry is that firms announcing on high-news days are intrinsically different from firms announcing on low-news days in ways the controls do not fully absorb.

To address this, the authors restrict the sample to firms that have at least 20% of their extreme-news-day announcements on both the high-news side and the low-news side. In this restricted sample, the key interaction coefficients remain similar in magnitude and remain statistically significant at about the 10% level.

Interpretation. This is an important check: the result is not driven only by one set of "always crowded" firms versus another set of "always quiet" firms.

4.6 Could busy days simply make earnings less informative?

A non-attention explanation would say that a firm's own earnings are just less informative on crowded days. If that were true, the *total* return response from day 0 to day 61 should be weaker on high-news days.

Table IV, Panel B rejects this view. The interaction in the total-response regression for $CAR[0, 61]$ is small and insignificant.

Interpretation. Distraction appears to change the *timing* of price adjustment, not the total informativeness of the earnings surprise.

4.7 Alternative abnormal-return constructions

The paper repeats its tests using beta-adjusted abnormal returns instead of characteristic-matched abnormal returns and gets very similar coefficients. This shows the findings are not an artifact of one particular abnormal-return benchmark.

4.8 What the paper can and cannot distinguish

The evidence is very persuasive that distraction matters. But the paper cannot fully separate every limited-attention mechanism.

In particular:

- it is consistent with *selective communication* (good news spreads more than bad news),
- it is also consistent with some forms of *naive extrapolation* or limited processing,
- and it does not pin down exactly which investor groups drive the reaction, even though the broader interpretation points toward limited attention in the market as a whole.

Interpretation. The asymmetry result—positive peer outcomes matter more than negative ones—leans especially strongly toward selective communication, but the paper remains appropriately cautious.

4.9 Relation to the literature

The paper speaks to three literatures at once:

- **Post-earnings-announcement drift:** it provides a direct limited-attention mechanism for why PEAD can be stronger in some situations than others.
- **Investor attention:** instead of relying on Fridays, media salience, or general market conditions as attention proxies, it directly measures competing news flow.
- **Behavioral asset pricing:** it offers a micro-structure for how underreaction can arise when investors' cognitive resources are temporarily crowded out.

5 Tables and figures

The paper's evidence proceeds in a tight sequence. First it shows that announcement days differ a lot in how much competing news they contain. Then it documents weaker immediate reactions and stronger delayed reactions on crowded days. After that it rules out calendar, leakage, and measurement stories, and finally it asks how economically important the distraction effect is and which kinds of news are most distracting.

5.1 Table I: what high-news days look like

Read:

- The average day has about 121 quarterly earnings announcements, but the distribution is very dispersed: the median is only 71, while the 90th percentile is 290.
- High-news-day announcements come disproportionately from larger firms, with more analyst following and higher institutional ownership.
- Reporting lag is strongly related to announcement crowding, which is why the paper devotes so much effort to controlling for it flexibly.

Economic message. Table I shows why a naive comparison of busy and quiet announcement days would be unconvincing. Busy days do not just have more news; they involve different kinds of firms and reporting patterns. The entire multivariate section is motivated by this table.

5.2 Table II and Figures 1–2: the basic distraction pattern is already visible in raw data

Read:

- Table II compares the spread between good-news and bad-news firms across low-news and high-news days.
- For immediate reactions, the good-minus-bad spread in $CAR[0,1]$ is 7.02% on low-news days and only 5.81% on high-news days.
- For post-announcement drift, the good-minus-bad spread in $CAR[2,61]$ is only 2.66% on low-news days but 7.18% on high-news days.
- Figure 1 shows the flatter announcement-return slope on high-news days; Figure 2 shows the steeper drift slope on high-news days.

Economic message. This is the raw visual heart of the paper. Competing announcements do not just coincide with lower announcement returns. They specifically weaken the *sensitivity* of immediate returns to earnings news and strengthen the delayed correction afterwards.

5.3 Table III: the baseline multivariate result

Read:

Table I
Sample Descriptive Statistics

Using quarterly earnings announcement dates from the CRSP-Compustat merged database and I/B/E/S for the period from January 1995 to December 2004, we calculate the total number of announcements on each day. In each calendar quarter, we sort quarterly earnings announcements during that quarter into deciles by the total number of announcements on the day of the announcement. Panel A reports the distribution of the daily total number of announcements. Panel B reports the average Size, B/M, *Earnings Surprise*, *Earnings Persistence*, institutional ownership (*IO*), *Earnings Volatility*, *Reporting Lag*, and the number of analysts following the firm (*# Analysts*) by the number-of-announcements deciles (*NRANK*), after deleting observations with missing information for any of these variables. See Section III.A for variable definitions. Panel C shows standardized regression coefficient estimates where *NRANK* is the dependent variable other variables in Panel B are independent variables. All variables are standardized to have mean zero and standard deviation of 1. Standard errors of coefficients are adjusted for heteroskedasticity and clustering by the day of announcement. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

Panel A: Distribution of Daily Number of Announcements

		Percentiles				
Mean	*SD*	P10	P25	Median	P75	P90
120.8	129.7	20	33	71	175	290

Panel B: Sample Characteristics by *NRANK*

NRANK	Size ($M)	B/M	Earnings Surprise	Earnings Persistence	*IO*	Earnings Volatility	Reporting Lag	# Analysts	Share Turnover
1	2,418	0.651	−0.56%	0.356	46.97%	2.15%	35.6	8.7	14.7%
2	2,966	0.723	−0.44%	0.342	46.91%	2.33%	31.8	9.0	14.5%
3	2,462	0.830	−0.33%	0.388	45.71%	2.22%	32.2	8.8	14.2%
4	2,319	0.801	−0.22%	0.391	45.43%	2.41%	34.3	8.6	14.5%
5	2,738	0.731	−0.25%	0.398	46.48%	2.24%	31.4	9.0	14.4%
6	3,490	0.735	−0.14%	0.431	48.45%	1.95%	27.1	9.9	14.9%
7	3,155	0.670	−0.16%	0.424	49.02%	2.06%	27.4	10.0	15.2%
8	3,373	0.656	−0.21%	0.426	49.45%	1.92%	25.6	10.2	15.3%
9	3,524	0.672	−0.05%	0.426	49.78%	1.98%	24.7	10.4	15.2%
10	3,315	0.780	−0.22%	0.413	49.33%	2.26%	26.0	10.1	15.5%
Difference (10–1)	897***	0.129	0.34%***	0.057***	2.36%***	0.11%	−9.5***	1.4***	0.7%***

Panel C: Regression Analysis

	Dependent Variable: *NRANK*
Size	−0.011***
	(0.004)
B/M	0.004
	(0.003)
Earnings Surprise	−0.006*
	(0.004)
Earnings Persistence	0.040***
	(0.004)
IO	−0.008
	(0.009)
Earnings Volatility	0.011***
	(0.004)
Reporting Lag	−0.246***
	(0.017)
# Analysts	0.023***
	(0.005)
Share Turnover	−0.005
	(0.005)
Constant	−0.000
	(0.030)
Observations	112,839
R^2	6.5%

Table II
Cumulative Abnormal Returns of Extreme Earnings Surprise Deciles by Number-of-Announcements Deciles

Using quarterly earnings announcements from January 1995 to December 2004, we calculate the average 2-day announcement cumulative abnormal returns (*CAR*[0,1]) and 60-day post-announcement cumulative abnormal returns (*CAR*[2,61]) for extreme earnings surprise deciles (*FE10*: good news, *FE1*: bad news) by the number-of-announcements deciles (*NRANK*). Earnings surprise and number-of-Announcements deciles are formed based on quarterly independent double sorts of quarterly earnings announcements by the corresponding forecast error and the number of quarterly earnings announcements on the day of announcement. The *p*-values are calculated using standard errors adjusted for heteroskedasticity and clustering by date. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Average *CAR*[0,1] for Earnings Surprise Deciles 1 and 10			Average *CAR*[2,61] for Earnings Earnings Surprise Deciles 1 and 10		
NRANK	*FE*1	*FE*10	*FE*10-*FE*1	*FE*1	*FE*10	*FE*10-*FE*1
1	−3.17%	3.86%	7.02%***	−0.70%	1.96%	2.66%*
2	−3.37%	3.51%	6.88%***	−1.57%	3.49%	5.05%***
3	−3.18%	3.63%	6.80%***	−1.56%	2.15%	3.71%**
4	−3.51%	2.54%	6.05%***	−4.47%	3.62%	8.09%***
5	−3.26%	2.28%	5.54%***	−2.32%	2.58%	4.90%***
6	−2.99%	3.27%	6.26%***	1.10%	3.93%	2.83%
7	−3.64%	2.78%	6.42%***	−2.86%	2.88%	5.74%***
8	−3.51%	2.79%	6.30%***	−2.61%	6.79%	9.40%***
9	−3.65%	2.82%	6.47%***	−1.21%	4.34%	5.54%***
10	−3.15%	2.66%	5.81%***	−2.73%	4.45%	7.18%***
Difference (10–1)	0.02%	−1.19%**	−1.21%*	−2.02%	2.49%	4.52%**

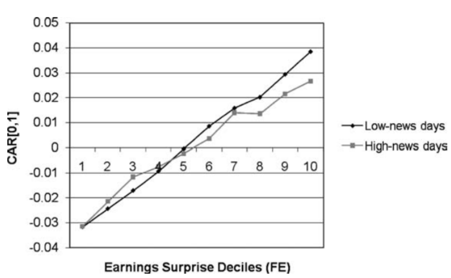

Figure 1. Market reactions to earnings news: *CAR*[0, 1]. Figure 1 shows the average 2-day announcement cumulative abnormal returns (*CAR*[0,1]) of quarterly earnings announcements against earnings surprise deciles (*FE* = 1: bad news, 10: good news) for announcements on high-news days (number-of-announcements Decile 10) and low-news days (number-of-announcements Decile 1). Earnings surprise and the number-of-announcements deciles are formed based on a quarterly independent double sort of quarterly earnings announcements by the corresponding forecast error and the number of quarterly earnings announcements on the day of the announcement.

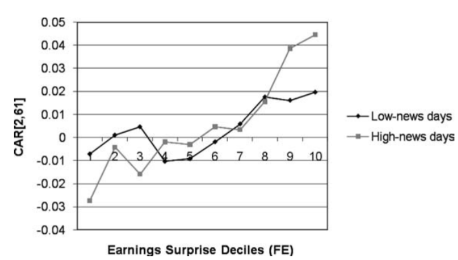

Figure 2. Post-earnings announcement drift: *CAR*[2, 61]. Figure 2 shows the average 60-day post-announcement cumulative abnormal returns (*CAR*[2, 61]) of quarterly earnings announcements against earnings surprise deciles (*FE* = 1: bad news, 10: good news) for announcements on high-news days (number-of-announcements decile 10) and low-news days (number-of-announcements Decile 1). Earnings surprise and the number-of-announcements deciles are formed based on a quarterly independent double sort of quarterly earnings announcements by the corresponding forecast error and the number of quarterly earnings announcements on the day of the announcement.

- With controls, the key interaction coefficient is -0.015 for $CAR[0,1]$ and $+0.049$ for $CAR[2,61]$.
- The paper translates these into economically meaningful terms: immediate return sensitivity to earnings news is about 13.3% lower on high-news days than on low-news days.
- The effect is not small compared with the standard determinants of earnings responses. For post-announcement drift, $NRANK$ is one of the most important variables in the regression.

Economic message. Table III matters because it shows the raw result survives when one controls for almost every standard confound in the earnings-response literature. The sign pattern is exactly what the distraction hypothesis predicts.

Table III

Market Reactions to Earnings News: Regression Analysis

Table III reports the multivariate tests of the effects of the number of announcements on the relation between announcement or post-announcement returns and earnings surprises. The dependent variable is indicated under each column heading. *FE* is the earnings surprise decile (*FE* = 1: lowest, 10: highest) and *NRANK* is the number-of-announcements decile, based on quarterly independent sorts by forecast error and the number of announcements on the day of announcement. Regressions (5) and (6) include observations in extreme earnings surprise deciles only (*FE* = 1 or 10); *FE*10 is an indicator variable for the top earnings deciles (*FE* = 10). Control variables include size and book-to-market deciles, log (1 + # *Analysts*), *Reporting Lag*, *Reporting Lag* squared and cubed, institutional ownership (*IO*), *Earnings Volatility, Earnings Persistence, Share Turnover,* and indicator variables for year, month, day of week, and Fama-French 10 industry classification. See Section III.A for variable definitions. All control variables are interacted with *FE* (*FE*10 for Regressions 5 and 6). Standard errors adjusted for heteroskedasticity and clustering by the day of announcement are in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

	(1) CAR[0,1]	(2) CAR[0,1]	(3) CAR[2,61]	(4) CAR[2,61]	(5) CAR[0,1]	(6) CAR[2,61]
FE	0.722***	1.028***	0.372***	0.979***		
	(0.022)	(0.098)	(0.074)	(0.322)		
FE × *NRANK*	−0.008**	−0.015***	0.034***	0.049**		
	(0.004)	(0.005)	(0.012)	(0.019)		
FE10					10.730***	9.869**
					(1.463)	(4.709)
FE10 × *NRANK*					−0.178**	0.511**
					(0.073)	(0.252)
Controls, interacted with *FE*		X		X	X	X
Constant	−3.817***	−5.386***	−2.092***	−7.429***	−3.479***	−7.695**
	(0.134)	(0.577)	(0.462)	(1.959)	(1.065)	(3.542)
Observations	112,839	112,839	112,839	112,839	22,203	22,203
R^2	5.1%	6.2%	0.3%	1.1%	9.9%	3.0%

Table IV

Market Reactions to Earnings News: Additional Analyses

Table IV reports additional multivariate tests of the effects of the number of announcements on the relation between returns and earnings surprises. *FE* is the earnings surprise decile (*FE* = 1: lowest, 10: highest) and *NRANK* is the number-of-announcements decile, based on quarterly independent sorts by forecast errors and the number of announcements on the day of announcement. Panel A reports the post-earnings announcement drift over 30-, 45-, 61-, 75-, and 90-day horizons, and also over days [3, 61]. In Panel B, Regressions (1) and (2) use beta-adjusted abnormal returns as dependent variables. Regressions (3) and (4) use the subset of firms for which at least 20% of their extreme-news-day announcements (defined as *NRANK* = 1, 2, 9, 10) occur on high-news days (*NRANK* = 9, 10), and at least 20% occur on low-news days (*NRANK* = 1, 2). Regression (5) shows the cumulative abnormal returns from the announcement day until day 61. Control variables include size and book-to-market deciles, log (1 + # *Analysts*), *Reporting Lag*, *Reporting Lag* squared and cubed, institutional ownership (*IO*), *Earnings Volatility, Earnings Persistence, Share Turnover,* and indicator variables for year, month, day of week, and Fama-French 10 industry classification. See Section III.A for variable definitions. All control variables are interacted with *FE*. Standard errors adjusted for heteroskedasticity and clustering by the day of announcement are in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

Panel A: Post-Earnings Announcement Drift over Different Horizons

	(1) *CAR*[2,30]	(2) *CAR*[2,45]	(3) *CAR*[2,61]	(4) *CAR*[2,75]	(5) *CAR*[2,90]	(6) *CAR*[3,61]
FE	0.508**	0.612**	0.979***	1.492***	1.487***	1.006***
	(0.206)	(0.265)	(0.322)	(0.374)	(0.403)	(0.315)
FE × *NRANK*	0.019	0.044***	0.049**	0.040*	0.034	0.047**
	(0.014)	(0.016)	(0.019)	(0.024)	(0.023)	(0.019)
Controls, interacted with *FE*	X	X	X	X	X	X
Constant	−3.945***	−4.758***	−7.429***	−9.508***	−9.806***	−7.590***
	(1.306)	(1.614)	(1.959)	(2.226)	(2.463)	(1.907)
Observations	112,839	112,839	112,839	112,525	112,077	112,839
R^2	0.8%	0.9%	1.1%	1.1%	1.2%	1.1%

Panel B: Further Robustness Checks

	Beta-Adjusted Returns		No Strong Preference Toward High or Low News Days		Total Response
	(1) *CAR*[0,1]	(2) *CAR*[2,61]	(3) *CAR*[0,1]	(4) *CAR*[2,61]	(5) *CAR*[0,61]
FE	1.010***	1.093***	1.203***	0.969*	2.028***
	(0.098)	(0.332)	(0.193)	(0.566)	(0.341)
FE × *NRANK*	−0.015***	0.052***	−0.017*	0.050*	0.030
	(0.005)	(0.020)	(0.009)	(0.029)	(0.021)
Controls, interacted with *FE*	X	X	X	X	X
Constant	−5.199***	−8.539***	−6.947***	−4.851	−12.877***
	(0.593)	(2.123)	(1.133)	(3.328)	(1.984)
Observations	112,225	112,225	38,650	38,650	112,839
R^2	6.3%	2.9%	6.1%	1.3%	2.3%

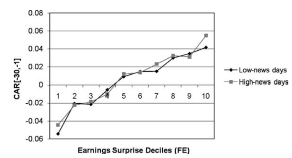

Figure 3. Preannouncement abnormal returns: *CAR*[−30, −1]. Figure 3 shows 30-day preannouncement cumulative abnormal returns (*CAR*[−30,−1]) of quarterly earnings announcements against earnings surprise deciles (*FE* = 1: bad news, 10: good news) for announcements on high-news days (number-of-announcements Decile 10) and low-news days (number-of-announcements Decile 1). Earnings surprise and the number-of-announcements deciles are formed based on a quarterly independent double sort of quarterly earnings announcements by the corresponding forecast error and the number of quarterly earnings announcements on the day of the announcement.

5.4 Table IV and Figure 3: the effect is gradual and there is no evidence of leakage

Read:

- Table IV shows that the extra drift after high-news-day announcements is insignificant over days $[2, 30]$, becomes significant by day 45, is strongest around day 61, and fades by day 90.
- The same table shows the result survives beta-adjusted abnormal returns, the restricted sample of firms that announce on both high- and low-news days, and alternative windows.
- The total-response regression $CAR[0, 61]$ does not show a significant distraction effect, which argues against the idea that earnings are simply less informative on high-news days.
- Figure 3 shows no differential preannouncement reaction on high-news versus low-news days.

Economic message. Table IV and Figure 3 jointly address the two most natural alternative explanations: bad timestamps and early information leakage. What remains looks like sluggish incorporation of relevant news, not a measurement artifact.

5.5 Table V and Table VI: seasonality matters, but actual same-day crowding still matters more

Read:

- Table V uses placebo measures based on the number of announcements on similar dates in the previous year.
- These placebo variables have predictive power when entered alone, which confirms that calendar seasonality is real.
- But once current-year $NRANK$ is included, the placebo interactions lose significance while the actual $FE \times NRANK$ interaction remains significant.
- Table VI shows that abnormal trading volume around the announcement is significantly lower on crowded days: the $NRANK$ coefficient is about -0.026 with full controls.

Economic message. Table V says the result is not just a Thanksgiving-or-third-Wednesday effect. Table VI then strengthens the attention story by showing that investors are not only pricing earnings less aggressively on crowded days; they are trading on them less aggressively as well.

Table V

Alternative Controls for Calendar Effects: Placebo Regressions

We include the number of announcements in the previous year in the regression analyses as alternative controls for calendar effects. *FE* is the earnings surprise decile (FE = 1: lowest, 10: highest) and *NRANK* is the number-of-announcements decile, based on quarterly independent sorts by forecast errors and the number of announcements on the day of announcement. *LNRANK1(placebo)* is the decile rank, based on last year's number of earnings announcements on the closest calendar day with the same weekday as the announcement date. *LNRANK2(placebo)* is the decile rank based on last year's number of earnings announcements on the day with the same weekday, same month, and same week of the month as the announcement date. Control variables include size and book-to-market deciles, log (1+ # *Analysts*), *Reporting Lag*, *Reporting Lag* squared and cubed, institutional ownership (*IO*), *Earnings Volatility, Earnings Persistence, Share Turnover,* and indicator variables for year, month, day of week, and Fama-French 10 industry classification. See Section III.A for variable definitions. All control variables are interacted with *FE*. Standard errors adjusted for heteroskedasticity and clustering by the day of announcement are in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

	(1) $CAR[0,1]$	(2) $CAR[0,1]$	(3) $CAR[0,1]$	(4) $CAR[0,1]$	(5) $CAR[2,61]$	(6) $CAR[2,61]$	(7) $CAR[2,61]$	(8) $CAR[2,61]$
FE	1.028***	1.008***	1.024***	1.026***	0.990***	1.067***	0.977***	0.992***
	(0.100)	(0.098)	(0.100)	(0.098)	(0.327)	(0.323)	(0.326)	(0.325)
$FE \times LNRANK1$(placebo)	−0.010*		0.002		0.036**		0.002	
	(0.005)		(0.008)		(0.018)		(0.024)	
$FE \times LNRANK2$(placebo)		−0.009*		0.002		0.026		−0.012
		(0.005)		(0.007)		(0.019)		(0.025)
$FE \times NRANK$			−0.017**	−0.016**			0.047*	0.057**
			(0.008)	(0.007)			(0.026)	(0.025)
Controls, interacted with FE	X	X	X	X	X	X	X	X
Constant	−5.411***	−5.307***	−5.417***	−5.375***	−7.647***	−7.838***	−7.574***	−7.387***
	(0.589)	(0.575)	(0.589)	(0.579)	(1.978)	(1.977)	(1.976)	(1.982)
Observations	112,839	112,839	112,839	112,839	112,839	112,839	112,839	112,839
R^2	6.2%	6.2%	6.2%	6.2%	1.1%	1.1%	1.1%	1.1%

Table VI

Trading Volume Response to Earnings News

We perform multivariate analysis of the effect of competing announcements on trading volume response to earnings news. Abnormal trading volume on a given day is defined as the log dollar trading volume on that day normalized by the average log dollar trading volume over days [−41, −11] of the announcement, and the dependent variable *VOL[0,1]* is the average abnormal trading volume over days [0,1] of the announcement. *AFE* is the absolute earnings surprise deciles and *NRANK* is the number-of-announcements deciles based on quarterly independent sorts by absolute forecast errors and the number of announcements on the day of announcement. Regressions (3) and (4) use indicator variables for each earnings surprise decile. Control variables include size and book-to-market deciles, log (1 + # *Analysts*), *Reporting Lag*, *Reporting Lag* squared and cubed, institutional ownership (*IO*), *Earnings Volatility, Earnings Persistence, Share Turnover,* and indicator variables for year, month, day of week, and Fama-French 10 industry classification. See Section III.A for variable definitions. In addition, we add market abnormal trading volume over days [0,1] of the announcement as an additional control variable, where market abnormal trading volume is defined as the average abnormal volume of all CRSP firms. Standard errors adjusted for heteroskedasticity and clustering by the day of announcement are in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

	(1) $VOL[0,1]$	(2) $VOL[0,1]$	(3) $VOL[0,1]$	(4) $VOL[0,1]$
$NRANK$	−0.020***	−0.026***	−0.022***	−0.025***
	(0.002)	(0.002)	(0.002)	(0.002)
AFE	0.019***	0.020***		
	(0.001)	(0.001)		
Indicator variables for FE deciles			X	X
Controls		X		X
Constant	0.582***	0.748***	0.622***	0.765***
	(0.014)	(0.040)	(0.017)	(0.040)
Observations	114,031	114,031	114,031	114,031
R^2	0.7%	4.6%	1.3%	5.2%

Table VII

Fama-French Alphas of Post-earnings Announcement Drift Portfolios

At the end of each month from March 1995 until December 2004, we independently sort stocks into 5 × 5 groups based on their most recent quarterly earnings surprises within the last 3 months (FE = 1–5) and the number of earnings announcements on the day of the earnings announcement ($NRANK$ = 1–5). We calculate equally weighted returns of the resulting 5 × 5 portfolios during the following month. Within each number-of-announcements rank (*NRANK*), we form a hedge portfolio that is long the good news portfolio ($FE = 5$) and short the bad news portfolio ($FE = 1$) to exploit post-earnings announcement drift. Alphas from time-series regressions of portfolio monthly returns (less the risk-free rate, except for the zero-cost hedge portfolios) on Fama-French three factors are reported with Newey–West standard errors with 12 lags in parentheses. The first row reports the alphas of the base post-earnings announcement drift portfolios, equally weighting firms in each earnings surprise quintile. The last row reports the alphas of hedge portfolios long the high-news-day portfolio ($NRANK = 5$) and short the low-news-day portfolio ($NRANK = 1$) within each earnings surprise quintile. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Earnings Surprise Quintile					
	$FE = 1$ (Bad News)	2	3	4	$FE = 5$ (Good News)	$FE5$-$FE1$ (Good−Bad)
All	−0.57	−0.53***	−0.12	0.04	0.75***	1.32***
	(0.36)	(0.20)	(0.14)	(0.14)	(0.26)	(0.26)
$NRANK1$	−0.21	−0.53**	−0.17	0.04	0.56**	0.77
	(0.42)	(0.22)	(0.21)	(0.21)	(0.24)	(0.47)
2	−1.10***	−0.56**	−0.04	−0.23	0.45	1.55***
	(0.37)	(0.24)	(0.21)	(0.18)	(0.28)	(0.26)
3	−0.51	−0.61***	0.02	−0.02	0.70**	1.22***
	(0.43)	(0.23)	(0.17)	(0.15)	(0.30)	(0.37)
4	−0.16	−0.56**	−0.26	0.32	0.94***	1.09***
	(0.50)	(0.25)	(0.16)	(0.20)	(0.34)	(0.40)
$NRANK5$	−0.60*	−0.44***	−0.15	0.06	1.04***	1.64***
	(0.34)	(0.17)	(0.14)	(0.17)	(0.34)	(0.31)
$NRANK5 - NRANK1$	−0.38	0.09	0.03	0.02	0.48*	0.86*
	(0.33)	(0.16)	(0.20)	(0.21)	(0.24)	(0.47)

5.6 Table VII: the trading strategy says the effect is economically meaningful

Read:

- The paper sorts stocks into portfolios by recent earnings surprise and by announcement-day crowding.
- The good-minus-bad hedge portfolio earns a three-factor alpha of 1.64% per month for high-news-day announcements.
- The corresponding alpha is only 0.77% and statistically insignificant for low-news-day announcements.

Economic message. Table VII turns the regression result into money. Distraction is not just statistically detectable; it meaningfully changes the profitability of a PEAD strategy.

5.7 Tables VIII and IX: what kinds of competing news and which firms matter most

Table VIII

Which Competing Announcements Are More Distracting?

We split competing announcements into two groups and compare the distraction effect of different types of competing announcements. We consider three different splits: (1) industry related vs. unrelated announcements, (2) big vs. small absolute earnings surprises, and 3) large vs. small firm announcements. *FE* is the earnings surprise decile (*FE* = 1: lowest, 10: highest) based on quarterly sorts by forecast errors. In the first set of regressions, we calculate the number of quarterly earnings announcements by the same industry firms ("related announcements") and the number of quarterly earnings announcements by firms in other industries ("unrelated announcements") using the Fama-French 10 industry classification, after excluding firms in Industry 10 ("Others"). *#RelatedNews* (*#UnrelatedNews*) is the decile rank of the number of related (unrelated) announcements (10: highest, 1: lowest) announcements based on quarterly sorts by the number of related (unrelated) announcements. In the second set of regressions, we split announcements in each calendar quarter into two groups by the absolute earnings surprises (small vs. big news). *#SmallNews* (*#BigNews*) is the decile rank of the number of competing announcements with small (big) absolute earnings surprise. In the third set, we split announcements into two groups by firm size (small vs. large firms). *#SmFirmNews* (*#LgFirmNews*) is the decile rank of the number of competing announcements by small (large) firms. Control variables include size and book-to-market deciles, log (1 + # *Analysts*), *Reporting Lag*, *Reporting Lag* squared and cubed, institutional ownership (*IO*), *Earnings Volatility, Earnings Persistence, Share Turnover,* and indicator variables for year, month, day of week, and Fama-French 10 industry classification. See Section III.A for variable definitions. All control variables are interacted with *FE*. Standard errors adjusted for heteroskedasticity and clustering by the day of announcement are in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Split by Industry Relatedness		Split by Absolute Earnings Surprise		Split by Firm Size	
	CAR[0,1]	*CAR*[2,61]	*CAR*[0,1]	*CAR*[2,61]	*CAR*[0,1]	*CAR*[2,61]
FE	1.046***	0.787**	1.006***	1.121***	0.974***	1.112***
	(0.111)	(0.368)	(0.105)	(0.330)	(0.105)	(0.324)
FE × *#RelatedNews*	−0.004	−0.002				
	(0.008)	(0.028)				
FE × *#UnrelatedNews*	−0.016**	0.056**				
	(0.007)	(0.026)				
FE × *#BigNews*			−0.017	0.099***		
			(0.010)	(0.030)		
FE × *#SmallNews*			0.001	−0.048		
			(0.011)	(0.030)		
FE × *#LgFirmNews*					0.003	−0.006
					(0.008)	(0.026)
FE × *#SmFirmNews*					−0.016***	0.048**
					(0.006)	(0.022)
Controls, interacted with *FE*	X	X	X	X	X	X
Constant	−5.702***	−7.731***	−5.321***	−8.429***	−5.101***	−8.594***
	(0.660)	(2.231)	(0.631)	(2.011)	(0.615)	(1.986)
Observations	89,095	89,095	112,839	112,839	112,839	112,839
R^2	6.4%	1.4%	6.2%	1.1%	6.2%	1.1%

Table IX

Variation in the Distraction Effect across Firm Size and the Sign of Earnings News

We examine how the distraction effect (*FE* × *NRANK*) varies with firm size and the sign of earnings news. *NRANK* is the number-of-announcements deciles based on quarterly sorts by the number of announcements on the day of the announcement. In Regressions (1) and (2), *FE* is the earnings surprise decile (*FE* = 1: lowest, 10: highest) based on quarterly sorts by forecast error. We split the sample into two groups by firm size (small vs. large) and create the indicator variable *LgFirm* (one for large firms, zero for small firms). Regressions (1) and (2) include additional variables, *FE* × *NRANK* × *LgFirm*, *NRANK* × *LgFirm*, *FE* × *LgFirm*, and *LgFirm*. Regressions (3) and (4) show estimates of piece-wise linear regressions, where we allow the return sensitivity to earnings news and the distraction effect to differ for positive surprises. *FE* is the earnings surprise *quantile*, where negative earnings surprises are ranked into the first five quantiles (1–5), zero surprises are assigned to quantile 6, and positive surprises are assigned to the top five quantiles, 7–11. FE_p is equal to Max(*FE*-6, 0). Control variables include size and book-to-market deciles, log (1 + # *Analysts*), *Reporting Lag*, *Reporting Lag* squared and cubed, institutional ownership (*IO*), *Earnings Volatility, Earnings Persistence, Share Turnover,* and indicator variables for year, month, day of week, and Fama-French 10 industry classification. See Section III.A for variable definitions. All control variables are interacted with *FE* and FE_p. Standard errors adjusted for heteroskedasticity and clustering by the day of announcement are in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

	(1) *CAR*[0,1]	(2) *CAR*[2,61]	(3) *CAR*[0,1]	(4) *CAR*[2,61]
FE	1.086***	0.886***	0.441**	1.859***
	(0.103)	(0.343)	(0.210)	(0.673)
FE_p			0.961***	−1.564
			(0.326)	(1.080)
FE × *NRANK*	−0.025***	0.069**	0.004	0.075**
	(0.008)	(0.027)	(0.011)	(0.037)
FE_p × *NRANK*			−0.033*	−0.057
			(0.018)	(0.060)
FE × *NRANK* × *LgFirm*	0.016**	−0.031		
	(0.007)	(0.026)		
Controls, interacted with *FE* and FE_p	X	X	X	X
Constant	−5.606***	−7.436***	−3.947***	−11.306***
	(0.612)	(2.101)	(1.041)	(3.403)
Observations	112,839	112,839	112,839	112,839
R^2	6.2%	1.1%	6.3%	1.5%

Read:

- Table VIII shows that industry-unrelated announcements are more distracting than related announcements. The coefficient on $FE \times \#UnrelatedNews$ is significant in both the immediate-return and drift regressions, while $FE \times \#RelatedNews$ is not.
- Competing announcements with large absolute earnings surprises are more distracting than those with small surprises. The drift coefficient on $FE \times \#BigNews$ is 0.099, highly significant, while the corresponding small-news term is insignificant.
- Small-firm competing announcements appear more distracting than large-firm announcements: $FE \times \#SmFirmNews$ is significant in both the immediate-return and drift regressions, while $FE \times \#LgFirmNews$ is not.
- Table IX provides weak evidence that distraction is stronger for small announcing firms. It also finds mixed evidence that distraction is somewhat stronger for positive earnings surprises than for negative ones.

Economic message. The comparative-statics section helps pin down the mechanism. Distraction is strongest when the competing news is the kind most likely to pull attention *away* from the focal firm rather than toward it—unrelated announcements and especially salient announcements.

6 Short summary

- The paper asks whether same-day earnings announcements by other firms distract investors from a firm's own earnings news.
- It measures distraction directly with the number of competing same-day earnings announcements.
- On crowded announcement days, the immediate price reaction to earnings news is weaker, post-announcement drift is stronger, and abnormal trading volume is lower.
- These effects survive rich controls, beta-adjusted abnormal returns, same-firm subsamples, alternative windows, and placebo tests based on prior-year announcement counts.
- The extra delayed reaction appears gradually, becoming significant around 45 trading days and peaking around 60 days.
- A PEAD trading strategy is much more profitable after high-news-day announcements than after low-news-day announcements.
- Unrelated competing news and big surprise announcements distract more than related or small competing news.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that competing same-day earnings announcements by other firms measurably distract investors. This distraction weakens the immediate reaction to a firm's own earnings surprise, lowers announcement-window trading volume, and intensifies subsequent post-earnings-announcement drift.

7.2 Economic intuition

The intuition is simple but powerful. Investors cannot process everything at once. When many firms announce earnings on the same day, some investors do not fully digest any one firm's announcement right away. Prices therefore move too little at first and continue to adjust later as the neglected information diffuses into the market.

7.3 Takeaway

The most important takeaway is that limited attention is not just a vague behavioral idea. In this paper it becomes a measurable state variable—the amount of competing same-day news. That makes the attention mechanism both testable and economically meaningful. More broadly, the paper suggests that underreaction anomalies can become stronger exactly when the information environment is the most crowded.